Breaking the Carlini Gauge: Black-Box Recovery of Transformer Output-Projection Columns via Third-Order Logit Moments

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Abstract

Carlini et al. [2024] showed a production LLM's output projection $\mathbf{W}_{\text{lm_head}}$ can be algebraically extracted from black-box logit queries via SVD — but only up to an invertible right-hand transform $\mathbf{A}$: the factorisation $\mathbf{z} = \mathbf{W}\mathbf{h} = (\mathbf{W}\mathbf{A})(\mathbf{A}^{-1}\mathbf{h})$ is a genuine second-order gauge ambiguity that no matrix-factorisation attack can cross. We break this ceiling. Using the third-order moment tensor $\mathbf{M}_3(\mathcal{Q}) = \mathbb{E}[\mathbf{z} \otimes \mathbf{e} \otimes \mathbf{e}] \in \mathbb{R}^{V \times d \times d}$, where $\mathbf{z}$ is the black-box logit vector and $\mathbf{e}$ is the *attacker-side* pooled input embedding (read off the public token-embedding table — no oracle access), a standard CP-ALS decomposition with Carlini-SVD warm-start empirically recovers $\mathbf{W}_{\text{lm_head}}$ *columns* on Qwen2.5-0.5B at top-5 aligned cosine $\mathbf{0.813}$ (mean 0.342 over 256 random-sampled columns, vs. a random-unit null of 0.163; gap $+0.650$). By Kruskal uniqueness, per-column identity is recoverable from third-order cross-moments in a way that no second-order factorisation can achieve — Carlini's SVD is mathematically limited to subspace identity ($\cos = 0.9999$ subspace, $\cos = 0$ per column by construction).

To explain why the same CP machinery does *not* recover deeper block-level weights at the same fidelity, we develop a matching *suffix observability* framework: the Gramian $\mathbf{G}(\mathcal{Q}) = \frac{1}{|\mathcal{Q}|} \sum_{\mathbf{x}} \mathbf{J}_{\mathbf{x}}^{\top} \mathbf{J}_{\mathbf{x}}$ on the symmetry-quotiented parameter space, its rank upper bound (Thm 7), and a depth-screening decomposition (Thm 8) showing the contribution of block $L - k$ to the Jacobian contracts by a uniform factor ρ^k set by the trailing suffix. These first-order bounds predict that any passive-query, algebraic, first-order attack on internal blocks remains at noise floor — which we verify empirically on Qwen2.5-0.5B and Llama-3.2-1B (pure-logits $\mathbf{W}_{\text{down}}$ recovery $\cos \approx 4 \times 10^{-5}$) and match against a $K_{\text{pos}} \times k$ sketched-Gramian rank-bound validation grid. The third-order attack complements the first-order barrier: it works exactly where the theory predicts a higher-order tensor has enough structure (the output projection sits one layer-norm away from the logits); it leaks statistically significant but weaker signal on four internal last-block matrices ($\mathbf{W}_V, \mathbf{W}_{\text{up}}, \mathbf{W}_Q, \mathbf{W}_{\text{gate}}$ above null margin); it fails on $\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$, where depth screening dominates.

The resulting attack setting is strictly black-box: only last-position logits and the public input-embedding table are used. We demonstrate the attack on a tied-embedding open-weight model (Qwen2.5-0.5B); untied architectures are discussed as an untested extension via Carlini's recovered pseudo-embedding $\hat{\mathbf{W}}_{\text{eff}}^{\top}$. The 0.813 top-5 result should be read as an *existence result at a demonstrated operating point* (4096 queries, CP rank 4864, 40 ALS iterations) rather than a full robustness or scaling characterization, which we leave for future work. The paper is thus a positive extraction result — the first per-column recovery of $\mathbf{W}_{\text{lm_head}}$ beyond Carlini's second-order ceiling — packaged with the observability theory that explains which internal weights remain out of reach for first-order attacks.

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1 Introduction

Large language models (LLMs) deployed as black-box APIs represent billions of dollars of investment in data curation, training compute, and algorithmic innovation. The security assumption underlying this deployment model is that API access — which returns only output logits or token probabilities — does not expose the model's internal parameters. But how valid is this assumption?

Recent work has begun to chip away at this barrier. Carlini et al. [2024] demonstrated that the output projection layer (`lm_head`) of production LLMs can be *algebraically extracted* from a modest number of API queries. Tramèr et al. [2016] showed that simpler models are vulnerable to equation-solving attacks. However, a fundamental question remains unanswered:

> *How deep into a transformer's architecture can weight recovery extend, and what governs the boundary between identifiable and non-identifiable layers?*

This question has profound implications. If internal transformer layers are identifiable from logits, the attack surface extends far beyond the output projection — potentially exposing proprietary architectural innovations, fine-tuning data through weight inspection, or enabling perfect model cloning that circumvents API rate limits and usage policies. Figure 1 provides a schematic overview of the problem and our findings. We note that our identifiability analysis is necessarily *query-dependent*: the Gramian characterizes identifiability with respect to a specific query distribution, and different (e.g., adversarially optimized) queries could in principle yield different observable subspaces.

This paper. We show that the third-order moment tensor

$$\mathbf{M}_3(\mathcal{Q}) = \frac{1}{|\mathcal{Q}|} \sum_{\mathbf{x} \in \mathcal{Q}} \mathbf{z}(\mathbf{x}) \otimes \mathbf{e}(\mathbf{x}) \otimes \mathbf{e}(\mathbf{x}) \in \mathbb{R}^{V \times d \times d}, \tag{1}$$

built entirely from the black-box logit vector $\mathbf{z}(\mathbf{x})$ and the attacker-computable pooled input embedding $\mathbf{e}(\mathbf{x})$, exposes per-column identity for $\mathbf{W}_{\text{lm_head}}$ via CP-ALS decomposition with Carlini-SVD warm-start. Kruskal uniqueness — the distinguishing property of third-order tensors absent from matrix factorisation — is what breaks the second-order gauge ambiguity that halts Carlini's SVD. Internal weight matrices $\mathbf{W}_V, \mathbf{W}_{\text{up}}, \mathbf{W}_Q, \mathbf{W}_{\text{gate}}$ leak statistically significant but weaker signal through the same mechanism; $\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$ remain at noise floor.

Why does the same CP machinery not recover deeper block weights at the same fidelity? We develop a matching **suffix observability** framework: the Gramian

$$\mathbf{G}(\mathcal{Q}) = \frac{1}{|\mathcal{Q}|} \sum_{\mathbf{x} \in \mathcal{Q}} \mathbf{J}_{\mathbf{x}}^{\top} \mathbf{J}_{\mathbf{x}}, \tag{2}$$

on the gauge-quotiented parameter space, a rank upper bound (Thm 7), and a depth-screening decomposition (Thm 8) showing that the contribution of block $L - k$ to the output Jacobian contracts by a uniform factor ρ^k set by the trailing suffix. The theory is *supporting*: it quantifies exactly which internal directions remain out of reach for first-order attacks, and predicts the depth-dependent signal decay we observe empirically.

Contributions.

- **Positive result (main contribution).** We introduce a third-order moment CP attack $\mathbf{M}_3 = \mathbb{E}[\mathbf{z} \otimes \mathbf{e} \otimes \mathbf{e}]$ with Carlini-SVD warm-start (§5) that empirically recovers per-column identity of $\mathbf{W}_{\text{lm_head}}$ on Qwen2.5-0.5B at top-5 aligned cosine 0.813 (mean 0.342) under a strict black-box threat model: only last-position logits and public input-embedding-table lookups are used. This exceeds what any second-order matrix-factorisation attack can achieve (Carlini's SVD: subspace 0.9999, per-column 0 by gauge).
- **Supporting theory.** We define the suffix observability Gramian on the symmetry-quotiented parameter space (§4), derive the continuous symmetry group of RMSNorm + gated-MLP transformers (§4.1), and prove three theorems: a first-order observability criterion (Thm 6), a Gramian rank upper bound (Thm 7), and a depth-screening decomposition (Thm 8). Together they explain *why* the same CP machinery does not recover deeper block-level weights with comparable fidelity: first-order Jacobian information about internal blocks is screened by the trailing suffix.

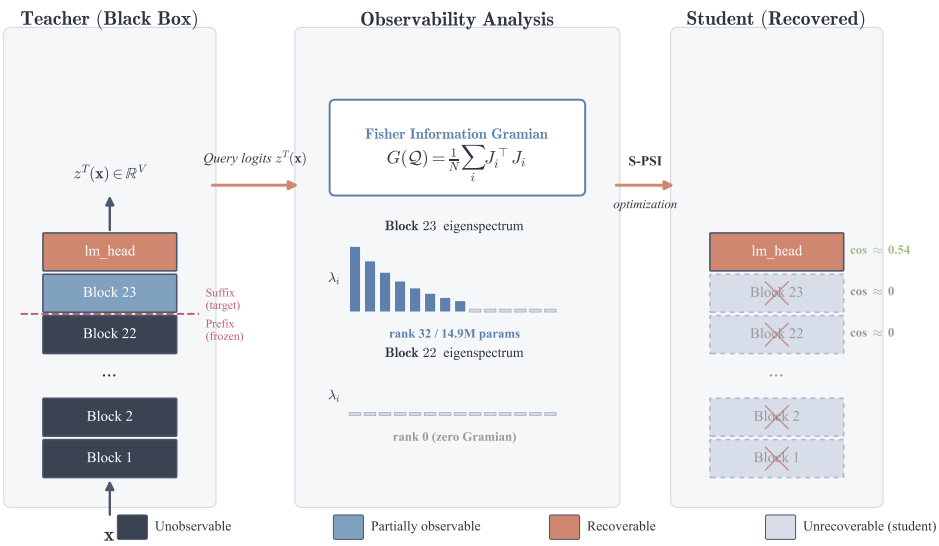

Figure 1: **Attack overview** (Qwen2.5-0.5B case study). The attacker queries a black-box teacher for last-position logits $\mathbf{z}(\mathbf{x})$, pools the public input-token-embedding table into $\mathbf{e}(\mathbf{x})$, forms the third-order moment tensor $\mathbf{M}_3 = \mathbb{E}[\mathbf{z} \otimes \mathbf{e} \otimes \mathbf{e}]$, and runs CP-ALS with Carlini-SVD warm-start. Output: per-column identity of $\mathbf{W}_{\text{lm_head}}$ at top-5 $\cos \approx 0.81$ via Kruskal uniqueness (impossible for any second-order matrix factorisation). Four internal last-block matrices ($\mathbf{W}_V, \mathbf{W}_{\text{up}}, \mathbf{W}_Q, \mathbf{W}_{\text{gate}}$) show weaker leakage above the null margin; $\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$ remain at noise floor, a pattern explained by the supporting observability theory (Thm 7–8).

- **Internal-layer leakage map.** Beyond $\mathbf{W}_{\text{lm_head}}$, the attack produces statistically significant but weaker signal on four last-block matrices ($\mathbf{W}_V, \mathbf{W}_{\text{up}}, \mathbf{W}_Q, \mathbf{W}_{\text{gate}}$, all above null margin 0.02), while $\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$ remain at null. We report this leakage map as a defender-facing diagnostic, not as full internal recovery.
- **Theorem validation.** A $K_{\text{pos}} \in \{32, 64, 128\} \times k \in \{64, 256\}$ sketched-Gramian rank grid on Qwen validates the rank upper bound; pure-logits algebraic reconstruction stays at noise floor on both Qwen and Llama-3.2-1B, corroborating the first-order observability barrier.
- We release code and experimental artifacts for reproducibility.

2 Related Work

Model extraction attacks. Model stealing has evolved from training surrogate models on query-response pairs [Tramèr et al., 2016] to functionally equivalent extraction [Jagielski et al., 2020, Orekondy et al., 2019]. Carlini et al. [2024] achieved a qualitative leap by showing that the final projection layer of production LLMs can be *algebraically* recovered — not merely approximated — from a small number of crafted API queries. Their attack exploits the affine structure of the last-layer mapping $\mathbf{z} = \mathbf{W}_{\text{lm_head}}\mathbf{h}_N + \mathbf{b}$. However, they explicitly note that extending extraction beyond the output layer remains an open problem, as deeper layers lack this simple algebraic structure. Our work directly addresses this open question.

Knowledge distillation and model compression. Knowledge distillation (KD) [Hinton et al., 2015, Gou et al., 2021] trains a student to match teacher outputs but does not aim for *parameter* recovery. The student learns a *functionally equivalent* mapping, potentially with entirely different internal representations. Our setting is fundamentally different: we seek to recover the teacher's actual weight matrices, not merely its input-output behavior. This distinction matters for security analysis, intellectual property claims, and understanding model identifiability.

Neural network identifiability. Theoretical identifiability of neural networks has a rich history. Sussmann [1992] showed that single hidden-layer networks with analytic activations are identifiable up to permutation symmetries. Bui and Lampert [2020] extended this to deeper networks under specific conditions. For transformers, Liu and Moitra [2025] recently studied identifiability of attention parameters. However, these works analyze identifiability *in principle* (given infinite, noiseless observations) rather than *in practice* from finite black-box queries. Our Gramian framework bridges this gap by quantifying identifiability as a function of the query set $\mathcal{Q}$.

Observability in control theory. The observability Gramian is a classical tool in linear systems theory [Kailath, 1980], characterizing which internal states can be inferred from output measurements. Extensions to nonlinear systems via empirical Gramians [Lall et al., 2002] compute $\mathbf{G} \approx \frac{1}{N}\sum_i \mathbf{J}_i^\top \mathbf{J}_i$ using Jacobians from perturbed trajectories. We adapt this framework to the novel setting of *parameter* observability in deep networks, where the "state" is the weight vector and "observations" are logit outputs across queries.

3 Problem Setup

Notation. Let T denote a frozen teacher transformer with L blocks, parameterized by $\boldsymbol{\theta}^T = (\boldsymbol{\theta}_1^T, \ldots, \boldsymbol{\theta}_L^T, \mathbf{W}_{\text{lm_head}}^T)$. The output logit mapping is $f_T : \mathbf{x} \mapsto \mathbf{z}^T(\mathbf{x}) \in \mathbb{R}^V$, where V is the vocabulary size. A student S has identical architecture but parameters $\boldsymbol{\theta}^S$.

Threat model. The attacker has:

1. **Black-box logit access**: can query $f_T(\mathbf{x})$ for arbitrary input sequences $\mathbf{x}$.
2. **Architecture knowledge**: knows the model class (e.g., Qwen2.5-0.5B) but not the weights.
3. **Query budget**: total of B API calls.

Suffix decomposition. We partition the transformer into a *prefix* (blocks $1, \ldots, L-K$) and a *suffix* (blocks $L-K+1, \ldots, L$ plus `lm_head`). The suffix parameters are $\boldsymbol{\theta}_{\text{suf}} \in \mathbb{R}^p$. For any input sequence $\mathbf{x}$ of length T, the prefix produces a boundary hidden-state sequence $\mathbf{H}_{L-K}(\mathbf{x}) \in \mathbb{R}^{T\times d}$, and the suffix maps $\mathbf{H}_{L-K} \mapsto \mathbf{Z} \in \mathbb{R}^{T\times V}$. For notational simplicity, we write $\mathbf{h}_{L-K}(\mathbf{x}) \in \mathbb{R}^d$ and $\mathbf{z}(\mathbf{x}) \in \mathbb{R}^V$ when treating a single token position; in practice, the Gramian is computed over the last K_{pos} positions, so each query contributes K_{pos} independent Jacobian rows. Note that the suffix's self-attention mechanism couples positions within a sequence: perturbing block ℓ's parameters changes the hidden state at all positions, and self-attention in downstream blocks propagates this effect across positions. Consequently, the per-query Jacobian factors through the full sequence state $\mathbf{H}_\ell \in \mathbb{R}^{Td}$, giving an effective per-query rank of up to Td (Theorem 7).

Experimental regimes. We study two complementary settings:

- **Boundary-state oracle** (upper bound on identifiability): The attacker receives the teacher's true boundary states $\mathbf{h}_{L-K}^T(\mathbf{x})$ for each query. This isolates suffix identifiability from prefix error propagation.
- **Pure-logits** (realistic attack): The student's prefix is random-initialized and frozen. Only logits $\mathbf{z}^T(\mathbf{x})$ are observed. This tests practical extraction limits.

The oracle regime establishes what *could* be recovered under perfect conditions; the pure-logits regime reveals the practical gap introduced by prefix mismatch.

4 Symmetry and Observability Theory

4.1 Continuous Symmetry Group of Transformers

Modern transformers with RMSNorm and gated MLPs possess continuous symmetries that render the parameterization non-unique even when the input-output mapping is fully determined. We derive the continuous symmetry group for architectures like Qwen and LLaMA, identifying four families of reparameterization invariances.

Definition 1 (Transformer symmetry group)**.** *The continuous symmetry group $\mathcal{G}$ of a transformer block with RMSNorm and SiLU-gated MLP consists of:*

1. ***RMSNorm scale absorption*** $(\mathbb{R}_{>0})^d$*: For each dimension $i \in [d]$, scaling the RMSNorm weight $g_i \mapsto \alpha_i g_i$ can be absorbed by the consumer weights $\mathbf{W} \mapsto \mathbf{W}\,\mathrm{diag}(\alpha_1^{-1}, \ldots, \alpha_d^{-1})$:*

$$\mathrm{RMSNorm}_{g'}(\mathbf{x}) = \frac{\mathbf{x}}{\mathit{rms}(\mathbf{x})} \odot g' = \frac{\mathbf{x}}{\mathit{rms}(\mathbf{x})} \odot (Dg), \quad \mathbf{W}'\mathbf{h}' = \mathbf{W}D^{-1}D\mathbf{h} = \mathbf{W}\mathbf{h}. \tag{3}$$

   Each block has two RMSNorm layers (`input_layernorm` feeding $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$, and `post_attention_layernorm` feeding $\mathbf{W}_{gate}, \mathbf{W}_{up}$), yielding $2d$ continuous degrees of freedom per block.

2. ***Gated MLP up/down neuron scaling*** $(\mathbb{R}_{>0})^{d_{ff}}$*: For the SiLU-gated FFN $\mathrm{FFN}(\mathbf{x}) = [\mathrm{SiLU}(\mathbf{W}_{gate}\mathbf{x}) \odot (\mathbf{W}_{up}\mathbf{x})]\mathbf{W}_{down}^\top$, each intermediate neuron n admits a scaling of the* linear *branch and the down projection:*

$$\mathbf{W}_{up}[n,:] \mapsto \alpha_n \mathbf{W}_{up}[n,:], \quad \mathbf{W}_{down}[:,n] \mapsto \alpha_n^{-1}\mathbf{W}_{down}[:,n], \tag{4}$$

   with $\mathbf{W}_{gate}$ unchanged. This is exact because the gate branch passes through the nonlinear SiLU *activation, which is* not *positively homogeneous (*$\mathrm{SiLU}(\alpha x) \neq \alpha\ \mathrm{SiLU}(x)$*), so only the linear up-branch and its consumer down-projection can absorb a scaling. This contributes d_{ff} continuous degrees of freedom per block.*

3. ***Attention V/O change-of-basis*** $\mathrm{GL}(d_{head})^{H_{kv}}$*: In GQA with H_{kv} shared KV groups, the value projection $\mathbf{W}_V^{(g)}$ is shared across all H/H_{kv} query heads in group g. For any invertible $S_g \in \mathrm{GL}(d_{head})$, the transformation $\mathbf{W}_V^{(g)} \mapsto S_g\mathbf{W}_V^{(g)}$ (and bias $\mathbf{b}_V^{(g)} \mapsto S_g\mathbf{b}_V^{(g)}$) together with $\mathbf{W}_O^{(h)} \mapsto \mathbf{W}_O^{(h)}S_g^{-1}$ for all heads h in group g preserves the output, since $\mathbf{W}_O^{(h)}S_g^{-1} \cdot S_g\mathbf{W}_V^{(g)} = \mathbf{W}_O^{(h)}\mathbf{W}_V^{(g)}$. This contributes $d_{head}^2 \cdot H_{kv}$ continuous degrees of freedom.*

4. ***Attention Q/K RoPE-commuting symmetry*** $(\mathbb{C}^*)^{(d_{head}/2)\cdot H_{kv}}$*: For each KV group g and each RoPE frequency pair $j \in [d_{head}/2]$, transformations $\mathbf{W}_Q^{(h)}[2j{:}2j{+}2,:] \mapsto R_j\mathbf{W}_Q^{(h)}[2j{:}2j{+}2,:]$ and $\mathbf{W}_K^{(g)}[2j{:}2j{+}2,:] \mapsto R_j^{-\top}\mathbf{W}_K^{(g)}[2j{:}2j{+}2,:]$ (and corresponding biases) for $R_j = a_jI_2 + b_jJ$ with $a_j^2 + b_j^2 > 0$, where $J = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, preserve all attention scores. This is because R_j commutes with every RoPE rotation in its 2×2 block. Each pair contributes 2 real degrees of freedom (isomorphic to $\mathbb{C}^*$), for a total of $d_{head} \cdot H_{kv}$ directions per block. The previously reported scalar Q/K scaling is the 1-D subgroup where all $R_j = \alpha I_2$.*

For Qwen2.5-0.5B with $d = 896$, $d_{\mathrm{ff}} = 4864$, $H = 14$ heads, $H_{\mathrm{kv}} = 2$ KV heads, and $d_{\mathrm{head}} = d/H = 64$, the per-block gauge dimensions are:

$$d_{g,\mathrm{block}} = \underbrace{2d}_{\text{RMSNorm}} + \underbrace{d_{\mathrm{ff}}}_{\text{MLP}} + \underbrace{d_{\mathrm{head}}^2 \cdot H_{\mathrm{kv}}}_{\text{V/O GL}} + \underbrace{d_{\mathrm{head}} \cdot H_{\mathrm{kv}}}_{\text{Q/K RoPE}} = 1792 + 4864 + 8192 + 128 = 14{,}976. \tag{5}$$

The attention gauge (8320 directions) is substantially larger than the scalar approximation (4 directions). Discrete symmetries (attention head permutations $\mathfrak{S}_H$ and FFN neuron permutations $\mathfrak{S}_{d_{\mathrm{ff}}}$) also exist but are handled post-hoc via Hungarian matching.

Remark 2 (Gauge dimensions at block vs. suffix level)**.** *The gauge directions form the Lie algebra $\mathfrak{g}$ of $\mathcal{G}$. We construct an orthonormal basis $\mathbf{U} \in \mathbb{R}^{p \times d_g^{suf}}$ for the tangent space at the current parameterization. Let*

$$d_g^{block} = 2d + d_{ff} + d_{head}^2 H_{kv} + d_{head}H_{kv}$$

denote the per-block gauge dimension. For a suffix of K blocks with a final RMSNorm and an untied `lm_head` *(our student setup), the full suffix gauge dimension is*

$$d_g^{suf} = K \cdot d_g^{block} + \underbrace{d}_{\text{final RMSNorm} \leftrightarrow \text{lm_head columns}}.$$

When we write $d_{g,\ell}$ for a single block ℓ (e.g. in Theorem 7) we mean d_g^{block}; when we write d_g in Theorem 6 and Appendix C we mean d_g^{suf} applied to whichever parameter set $\boldsymbol{\theta}$ the statement is about.

Our current code implementation includes only the RMSNorm and MLP gauge ($2d + d_{ff} = 6656$ per block plus d for the final norm); the 8320 attention gauge directions per block are omitted from the explicit projection. However, by Proposition 4, these directions lie in $\ker(\mathbf{G})$ regardless of whether they are explicitly projected out, so the Gramian eigenspectrum is unaffected. The impact is on the interpretation*: the non-gauge parameter count $p - d_g^{suf}$ is smaller than what the code's projection target suggests. For Qwen2.5-0.5B, the attention gauge adds $\sim 0.06\%$ to the per-block gauge fraction, which does not materially change any conclusion.*

4.2 The Suffix Observability Gramian

Definition 3 (Suffix observability Gramian)**.** *Given a query set $\mathcal{Q} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ and a student parameterization $\boldsymbol{\theta}_{suf}$, the suffix observability Gramian is:*

$$\mathbf{G}(\mathcal{Q}) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{J}_i^\top \mathbf{J}_i \in \mathbb{R}^{p\times p}, \quad \textit{where } \mathbf{J}_i = \frac{\partial \mathbf{Z}(\mathbf{x}_i)}{\partial \boldsymbol{\theta}_{suf}} \in \mathbb{R}^{K_{pos}V\times p}. \tag{6}$$

Here $\mathbf{Z}(\mathbf{x}_i) \in \mathbb{R}^{K_{pos}\times V}$ denotes the logits at the last K_{pos} token positions (flattened to $\mathbb{R}^{K_{pos}V}$ for the Jacobian). When $K_{pos} = 1$, this reduces to $\mathbf{J}_i \in \mathbb{R}^{V\times p}$.

$\mathbf{G}(\mathcal{Q})$ is a symmetric positive semi-definite matrix. Its null space contains parameter directions that produce *zero first-order logit change* across all queries in $\mathcal{Q}$ at the current parameterization — these directions are locally first-order unobservable (higher-order or nonlocal identifiability is not addressed by the Gramian).

Proposition 4 (Gauge null-space)**.** *Let $\mathfrak{g}$ be the Lie algebra of continuous symmetries with basis $\mathbf{U}$. Then $\mathbf{G}(\mathcal{Q})\mathbf{U} = \mathbf{0}$, i.e., all gauge directions lie in the null space of $\mathbf{G}$ regardless of $\mathcal{Q}$.*

Proof. By definition, a continuous symmetry transforms $\boldsymbol{\theta} \mapsto \boldsymbol{\theta} + \epsilon\mathbf{u}$ for $\mathbf{u} \in \mathfrak{g}$ without changing the output: $\mathbf{z}(\mathbf{x}; \boldsymbol{\theta} + \epsilon\mathbf{u}) = \mathbf{z}(\mathbf{x}; \boldsymbol{\theta})$ for all $\mathbf{x}$. Differentiating at $\epsilon = 0$: $\mathbf{J}_{\mathbf{x}}\mathbf{u} = 0$ for all $\mathbf{x}$, hence $\mathbf{G}\mathbf{u} = 0$. □

This motivates working on the *gauge-quotiented* parameter space.

Definition 5 (Projected observability Gramian)**.** *Let $P_\perp = \mathbf{I} - \mathbf{U}\mathbf{U}^\top$ be the projection onto the gauge-orthogonal complement. The projected Gramian is:*

$$\mathbf{G}_\perp(\mathcal{Q}) = P_\perp \mathbf{G}(\mathcal{Q}) P_\perp. \tag{7}$$

Theorem 6 (Local first-order observability criterion)**.** *A parameter direction $\mathbf{v} \perp \mathfrak{g}$ is* locally first-order observable *(i.e., detectable) from $\mathcal{Q}$ if and only if $\mathbf{v}^\top \mathbf{G}_\perp(\mathcal{Q})\mathbf{v} > 0$: there exists a query $\mathbf{x}_i \in \mathcal{Q}$ such that $\mathbf{J}_i\mathbf{v} \neq 0$. The suffix parameters are* locally first-order identifiable *(up to gauge symmetries) from $\mathcal{Q}$ — meaning every non-gauge direction is uniquely distinguishable from all others at first order — if and only if $\operatorname{rank}(\mathbf{G}_\perp(\mathcal{Q})) = p - d_g^{suf}$, where d_g^{suf} is the suffix-level gauge dimension of Remark 2. Note that per-direction observability ($\mathbf{v}^\top \mathbf{G}_\perp \mathbf{v} > 0$) is necessary but not sufficient for unique identifiability of $\mathbf{v}$; the full-rank condition is needed for the latter. Neither condition implies global identifiability, which additionally requires analysis of nonlinear effects.*

4.3 Sketched Computation

The full Gramian $\mathbf{G} \in \mathbb{R}^{p\times p}$ is prohibitively large (e.g., $p \approx 10^7$ for a single block). We employ a sketched estimator: draw k orthonormal probe directions $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_k] \in \mathbb{R}^{p\times k}$ (after gauge projection) and compute the $k \times k$ *sketched Gramian*:

$$\tilde{\mathbf{G}} = \mathbf{V}^\top \mathbf{G} \mathbf{V} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{S}_i^\top \mathbf{S}_i, \quad \text{where } \mathbf{S}_i = [\mathbf{J}_i\mathbf{v}_1, \ldots, \mathbf{J}_i\mathbf{v}_k] \in \mathbb{R}^{K_{\text{pos}}V\times k}. \tag{8}$$

Each column $\mathbf{J}_i\mathbf{v}_j$ is computed efficiently via a single forward-mode JVP (Jacobian-vector product), avoiding the full Jacobian materialization. For k probes and N queries, this requires kN JVP evaluations.

4.4 Rank Upper Bound and Depth Screening

The following results explain the low-rank structure observed empirically.

Theorem 7 (Gramian rank upper bound for suffix block ℓ)**.** *Let block ℓ have parameter count p_ℓ, gauge dimension $d_{g,\ell}$, and hidden dimension d. Let T denote the sequence length. Since self-attention in blocks $\ell+1, \ldots, L$ couples token positions, the per-query logit Jacobian factors through the* full sequence *hidden state $\mathbf{H}_\ell \in \mathbb{R}^{T\times d}$:*

$$\mathbf{J}_{\mathbf{x}_i,\ell} = \underbrace{\frac{\partial \mathbf{Z}(\mathbf{x}_i)}{\partial \mathbf{H}_\ell}}_{\in \mathbb{R}^{K_{pos}V\times Td}} \cdot \underbrace{\frac{\partial \mathbf{H}_\ell}{\partial \boldsymbol{\theta}_\ell}}_{\in \mathbb{R}^{Td\times p_\ell}} \in \mathbb{R}^{K_{pos}V\times p_\ell}, \tag{9}$$

where $\mathbf{H}_\ell$ is flattened to $\mathbb{R}^{Td}$. There are two bottlenecks: the Td-dimensional full-sequence hidden state, and the $K_{pos}d$-dimensional observed output (since the logits at each of K_{pos} positions factor through a d-dimensional final hidden state):

$$\operatorname{rank}\big(\mathbf{G}_\perp^{(\ell)}(\mathcal{Q})\big) \le \min\Big(p_\ell - d_{g,\ell},\ \ K_{pos}\cdot d\cdot|\mathcal{Q}|,\ \ Td\cdot|\mathcal{Q}|\Big). \tag{10}$$

The $K_{pos}d$ bound is tighter when $K_{pos} < T$ (the typical experimental regime); Td applies only when the full sequence is observed ($K_{pos} = T$). For Qwen2.5-0.5B with $d = 896$, $K_{pos} = 8$, $T = 128$, and $p_\ell = 14.9M$: per-query rank $\le K_{pos}d = 7{,}168$; with $N = 2048$ queries, $\operatorname{rank} \le 7{,}168 \times 2{,}048 \approx 14.7M \approx p_\ell - d_{g,\ell}$, making the observation-count bound and the parameter-count bound comparable.

Proof. The Gramian $\mathbf{G}^{(\ell)} = \frac{1}{N}\sum_i \mathbf{J}_{\mathbf{x}_i,\ell}^\top \mathbf{J}_{\mathbf{x}_i,\ell}$ has $\operatorname{rank}(\mathbf{G}^{(\ell)}) \le \operatorname{rank}([\mathbf{J}_{\mathbf{x}_1,\ell};\ldots;\mathbf{J}_{\mathbf{x}_N,\ell}])$. Each $\mathbf{J}_{\mathbf{x}_i,\ell}$ factors through $\mathbf{H}_\ell \in \mathbb{R}^{Td}$ via (9), giving $\operatorname{rank}(\mathbf{J}_{\mathbf{x}_i,\ell}) \le Td$. A tighter per-query bound follows from the output side: let $\tilde{\mathbf{h}}_L[t_j] \in \mathbb{R}^d$ denote the post-final-RMSNorm representation at observed position t_j, so that $\mathbf{z}(\mathbf{x}_i)[t_j] = \mathbf{W}_{\text{lm}}\tilde{\mathbf{h}}_L[t_j]$ (the final RMSNorm only reweights components of $\mathbf{h}_L$, and does not increase rank). Each observed position therefore factors through a d-dimensional pre-logit state, so $\operatorname{rank}(\partial\mathbf{Z}/\partial\mathbf{H}_\ell) \le \sum_{j=1}^{K_{\text{pos}}} d = K_{\text{pos}}d$. Combined: $\operatorname{rank}(\mathbf{J}_{\mathbf{x}_i,\ell}) \le \min(K_{\text{pos}}d, Td)$. Stacking N queries: $\operatorname{rank} \le N\cdot\min(K_{\text{pos}}d, Td)$. Since all gauge directions lie in $\ker(\mathbf{G})$ by Proposition 4, the rank is further bounded by $p_\ell - d_{g,\ell}$. □

Theorem 8 (Depth screening)**.** *For block $\ell < L$, the Gramian decomposes using the factorization* (9)*:*

$$\mathbf{G}^{(\ell)}(\mathcal{Q}) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{B}_i^\top \cdot \mathcal{A}_i^\top \mathbf{L}_i^\top \mathbf{L}_i \mathcal{A}_i \cdot \mathcal{B}_i, \tag{11}$$

where $\mathcal{A}_i = \partial F_{\ell+1:L}/\partial\mathbf{H}_\ell|_{\mathbf{x}_i} \in \mathbb{R}^{Td\times Td}$ is the full sequence-level downstream Jacobian (including cross-position coupling from self-attention), $\mathcal{B}_i = \partial\mathbf{H}_\ell/\partial\boldsymbol{\theta}_\ell|_{\mathbf{x}_i} \in \mathbb{R}^{Td\times p_\ell}$ is block ℓ's parameter Jacobian, and $\mathbf{L}_i := \partial\mathbf{Z}(\mathbf{x}_i)/\partial\mathbf{H}_L|_{\mathbf{x}_i} \in \mathbb{R}^{K_{pos}V\times Td}$ is the query-dependent output Jacobian that folds in the final RMSNorm (whose scaling depends on $\operatorname{rms}(\mathbf{h}_L[t])$ per position), selection of the last K_{pos} positions, and the fixed $\mathbf{W}_{lm}$. All three factors $\mathcal{B}_i$, $\mathcal{A}_i$, $\mathbf{L}_i$ are query-dependent and cannot be factored out of the sum. The projected Gramian is $\mathbf{G}_\perp^{(\ell)} = P_\perp \mathbf{G}^{(\ell)} P_\perp$.

The maximum singular value of each summand is bounded by $\sigma_{\max}(\mathbf{L}_i\mathcal{A}_i\mathcal{B}_i)^2 \le \sigma_{\max}(\mathbf{L}_i)^2 \cdot \sigma_{\max}(\mathcal{A}_i)^2 \cdot \sigma_{\max}(\mathcal{B}_i)^2$. Write $\mathcal{A}_i = \prod_{j=\ell+1}^{L} D_{i,j}$ where $D_{i,j} := \partial\mathbf{H}_j/\partial\mathbf{H}_{j-1}|_{\mathbf{x}_i}$ is the sequence-level Jacobian of block j at query $\mathbf{x}_i$; then $\sigma_{\max}(\mathcal{A}_i) \le \prod_{j=\ell+1}^{L}\sigma_{\max}(D_{i,j})$. If there exists $\rho < 1$ such that $\sup_{i\in[N],\, j>\ell}\sigma_{\max}(D_{i,j}) \le \rho$ (i.e., the contraction is uniform over both queries and downstream blocks), then $\sigma_{\max}(\mathcal{A}_i) \le \rho^{L-\ell}$ for every i, giving exponential decay with depth. Under merely $\sigma_{\max}(D_{i,j}) < 1$ pointwise, the product may decay subexponentially. This provides a qualitative decomposition of depth-dependent Gramian spectra; the actual screening depends on the specific Jacobians at the current parameterization, and whether residual connections produce contractive or expansive $D_{i,j}$ is an empirical question not predicted by the theorem.

Proof. The decomposition (11) follows from the chain rule applied to the composite map $\boldsymbol{\theta}_\ell \mapsto \mathbf{H}_\ell \mapsto \mathbf{H}_L \mapsto \mathbf{Z}$, giving $\mathbf{J}_{\mathbf{x}_i,\ell} = \mathbf{L}_i\mathcal{A}_i\mathcal{B}_i$; hence $\mathbf{J}_{\mathbf{x}_i,\ell}^\top\mathbf{J}_{\mathbf{x}_i,\ell} = \mathcal{B}_i^\top\mathcal{A}_i^\top\mathbf{L}_i^\top\mathbf{L}_i\mathcal{A}_i\mathcal{B}_i$ and averaging

over i yields the stated Gramian form. Projection with $P_\perp$ on either side commutes with the sum, giving $\mathbf{G}_\perp^{(\ell)}$. The spectral bound $\sigma_{\max}(\mathbf{L}_i\mathcal{A}_i\mathcal{B}_i) \le \sigma_{\max}(\mathbf{L}_i) \cdot \sigma_{\max}(\mathcal{A}_i) \cdot \sigma_{\max}(\mathcal{B}_i)$ is submultiplicativity of the operator norm on matrix products. The factorization $\mathcal{A}_i = \prod_{j=\ell+1}^{L} D_{i,j}$ follows by iterating the chain rule across consecutive blocks, and submultiplicativity again gives $\sigma_{\max}(\mathcal{A}_i) \le \prod_{j=\ell+1}^{L} \sigma_{\max}(D_{i,j})$. Under the uniform contraction hypothesis $\sup_{i,j} \sigma_{\max}(D_{i,j}) \le \rho < 1$, the product is at most $\rho^{L-\ell}$. The subexponential case under pointwise-only contraction follows from standard sequence inequalities (e.g., AM-GM-style arguments on the log of singular values) and depends on how widely $\sigma_{\max}(D_{i,j})$ varies across j; we state this qualitatively because the paper uses the decomposition descriptively, not to certify any specific rate. □

Remark 9 (Dimensional argument for query complexity)**.** *A heuristic counting argument motivates the difficulty of full parameter recovery. The $k \times k$ sketched Gramian $\tilde{\mathbf{G}} = \mathbf{V}^\top \mathbf{G} \mathbf{V}$ captures at most k observable directions per evaluation. With N queries and k probes, each sketched Gramian step probes k directions out of $p_\ell - d_{g,\ell}$ non-gauge parameters. Covering all non-gauge directions would require at least $\lceil (p_\ell - d_{g,\ell})/k \rceil$ linearly independent probe sets. For Block 23 with $k = 128$ and $p_{23} - d_{g,23} \approx 14.9M$, this gives $\gtrsim 116{,}000$ independent sketches — far beyond what a single Gramian evaluation provides. This is a dimensional counting heuristic, not a formal information-theoretic bound (which would require specifying a noise model, regularity conditions, and estimator); it nonetheless quantifies the gap between the sketched subspace and the full parameter space.*

5 Method: Third-Order Moment CP Attack

The theory of Section 4 characterizes what *first-order* Jacobian information can and cannot expose, and Carlini et al. [2024]'s SVD attack already saturates the practical ceiling at second order: it recovers $\mathbf{W}_{\text{lm_head}}$ up to an invertible right-hand transform $\mathbf{A}$ (the factorization $\mathbf{Z} = \mathbf{W}\mathbf{H} = (\mathbf{W}\mathbf{A})(\mathbf{A}^{-1}\mathbf{H})$ is a genuine gauge ambiguity of matrix factorization). To recover per-column identity we move to third order.

Moment tensor. Let $\mathbf{z}(\mathbf{x}) \in \mathbb{R}^V$ be the last-position logit vector (black-box observable) and let $\mathbf{e}(\mathbf{x}) \in \mathbb{R}^d$ be the *attacker-computable* pooled input-token embedding

$$\mathbf{e}(\mathbf{x}) \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbf{W}_{\text{emb}}[x_t, :] \qquad \text{(mean-pooling over the input token sequence)}, \tag{12}$$

where $\mathbf{x} = (x_1, \ldots, x_T)$ are the input token ids and $\mathbf{W}_{\text{emb}} \in \mathbb{R}^{V \times d}$ is the *public* token-embedding matrix. For Qwen2.5-0.5B's tied-embedding architecture, $\mathbf{W}_{\text{emb}} = \mathbf{W}_{\text{lm_head}}^\top$ is publicly available from the model hub; for an untied model, the attacker can substitute the Carlini-SVD-recovered pseudo-embedding $\hat{\mathbf{W}}_{\text{eff}}^\top$ (an extension we argue but do not evaluate in this paper). Either way, computing $\mathbf{e}(\mathbf{x})$ requires no forward pass through the teacher and no intermediate hidden-state access — only the public embedding table and the input token ids. We form the third-order cross moment

$$\mathbf{M}_3(\mathcal{Q}) \;=\; \frac{1}{|\mathcal{Q}|}\sum_{\mathbf{x}\in\mathcal{Q}} \mathbf{z}(\mathbf{x}) \otimes \mathbf{e}(\mathbf{x}) \otimes \mathbf{e}(\mathbf{x}) \;\in\; \mathbb{R}^{V \times d \times d}, \tag{13}$$

whose entries are $\mathbf{M}_3[v, i, j] = \mathbb{E}[z_v \cdot e_i \cdot e_j]$. This is an asymmetric variant of the classical method-of-moments tensor studied by Anandkumar et al. [2014] for latent variable models: in our setting the first mode indexes vocabulary (logit coordinates z_v) while the two remaining modes share the hidden dimension of the public embedding table (input features e_i, e_j).

CP decomposition. We seek a rank-r decomposition

$$\mathbf{M}_3 \;\approx\; \sum_{j=1}^{r} \lambda_j\, \mathbf{a}_j \otimes \mathbf{b}_j \otimes \mathbf{c}_j, \qquad \mathbf{a}_j \in \mathbb{R}^V,\ \mathbf{b}_j, \mathbf{c}_j \in \mathbb{R}^d, \tag{14}$$

where the factors $\{\mathbf{a}_j\}_{j=1}^r$ are expected to align (up to permutation and sign/scale) with columns of $\mathbf{W}_{\text{lm_head}}$ that have non-trivial third-moment signature. We use alternating least squares (ALS) with Tikhonov-regularised Gram updates, running for a fixed iteration budget with tolerance 10^{-5} on relative fit.

Carlini SVD warm-start. Random ALS initialization often converges to local optima that mix multiple true factors. We therefore warm-start the first mode from the SVD-recovered subspace: we first run Carlini's SVD on the $|\mathcal{Q}| \times V$ logit matrix to obtain $\hat{\mathbf{W}}_{\text{eff}}$ (rotated $\mathbf{W}_{\text{lm_head}} \cdot \text{diag}(\mathbf{g}_{\text{final}})$), then initialize $[\mathbf{a}_1, \ldots, \mathbf{a}_d]$ with its orthonormal right singular vectors transposed back to the vocabulary space. Remaining factors are random. Empirically, this both accelerates ALS convergence and concentrates signal on per-column matches: CP uniqueness (Kruskal conditions on d-mode rank of the factor matrices) then distinguishes permutation/scaling ambiguity from the deeper rotational ambiguity that halts Carlini's second-order attack.

Evaluation via top-k aligned cosine. We report two metrics per weight matrix. First, *aligned top-5 mean cosine*: for each recovered factor $\mathbf{a}_j$, find the best-matching teacher column $\mathbf{w}_m$ by maximum absolute cosine, then average over the top 5 factors. Second, *mean aligned cosine* over 256 sub-sampled teacher columns (for large matrices like $\mathbf{W}_{\text{lm_head}}$ with $V = 151{,}936$). A random-sign, random-unit-norm null baseline is computed with the same evaluator; a matrix is declared *leaked* if its real top-5 exceeds the null by at least 0.02 (null margin).

Relation to Finlayson et al. 2024. Our attack realises and empirically evaluates a proposal sketched in Finlayson et al. [2024, §7.3] "recovering the softmax matrix from outputs". Finlayson et al. describe an SVD-based approach that does *not* cross the gauge barrier; our tensor formulation does (see §2).

Relation to Golowich et al. 2025. Golowich et al. [2025], Golowich [2025] study low-rank surrogates (ISAN) that match the output distribution in TV distance without recovering transformer weights. Our attack is orthogonal: we recover *actual weights* per column rather than distribution-matching surrogates.

6 Experiments

6.1 Setup

Model. We use Qwen2.5-0.5B (24 blocks, $d = 896$, $d_{\text{ff}} = 4864$, 14 attention heads with 2 KV heads (GQA), RMSNorm, SiLU-gated MLP, $V = 151{,}936$, and tied input/output embeddings in the teacher). The teacher is the pre-trained model. The student has identical architecture but with the `lm_head` explicitly *untied* from the input embedding, so that the output projection can be optimized as a free suffix parameter (the continuous symmetries of Definition 1 therefore act on the student's independent `lm_head`, not on the shared embedding).

Query pool. WikiText-103 [Merity et al., 2016] with 80/20 train/held-out split. Primary diagnostic experiments (Table 3) use a pool of 4096 sequences at maximum length 192 tokens; the $K_{\text{pos}} \times k$ Gramian sweep (Table 2, Figure 2) uses 2048 sequences at 128 tokens to enable direct Jacobian computation on full-sequence hidden states within GPU memory.

Hardware. 4× NVIDIA A100-80GB. Each experiment runs on a single GPU.

Evaluation metrics. (i) Sketched Gramian effective rank and condition number; (ii) per-matrix cosine of student to teacher after discrete-symmetry alignment (Hungarian matching for heads/neurons); (iii) for the third-order moment attack, *aligned top-5 mean cosine* and *aligned mean cosine* per weight matrix against a random-unit-norm null (margin 0.02).

6.2 Moments CP Attack: $\mathbf{W}_{\text{lm_head}}$ Column Recovery (Headline Result)

We run the third-order moment CP attack of Section 5 on Qwen2.5-0.5B with the last block (block 23) as the target. The attacker uses 4,096 random-token sequences of length 128 as queries, forms $\mathbf{M}_3 \in \mathbb{R}^{894 \times 896 \times 896}$ after Carlini-SVD dimensionality reduction of the vocabulary mode ($\hat{d} = 894$), and runs ALS for 40 iterations at rank $r = 4864$ (matching d_{ff}).

Output projection recovered per-column. Table 1 reports aligned cosine similarity of recovered CP factors against teacher weight matrices. The headline is $\mathbf{W}_{\text{lm_head}}$: top-5 mean $\cos = 0.81$

Table 1: **Third-order moment CP attack** on Qwen2.5-0.5B last block (Carlini-SVD warm-start, 4096 random-token queries, rank $r = 4864$, 40 ALS iterations). Top-5 mean and max are over best-matching teacher columns/rows; mean is over 256 sub-sampled teacher columns; null is a random-unit-norm baseline with the same evaluator. Bold indicates leakage signal above null margin (0.02).

Matrix	mean	top5	max	null top5	gap
$\mathbf{W}_{\text{lm_head}}$ (cols)	**0.342**	**0.813**	**0.824**	0.163	**+0.650**
$\mathbf{W}_V$ (rows)	0.097	**0.231**	**0.257**	0.158	**+0.073**
$\mathbf{W}_{\text{up}}$ (rows)	0.077	**0.208**	**0.278**	0.159	**+0.048**
$\mathbf{W}_Q$ (rows)	0.087	**0.200**	0.239	0.158	**+0.041**
$\mathbf{W}_{\text{gate}}$ (rows)	0.082	**0.198**	0.228	0.163	**+0.035**
$\mathbf{W}_K$ (rows)	0.089	0.165	0.193	0.158	+0.008
$\mathbf{W}_O$ (cols)	0.076	0.168	0.207	0.157	+0.010
$\mathbf{W}_{\text{down}}$ (cols)	0.064	0.139	0.171	0.158	−0.020

(vs. null 0.16, gap +0.65), mean cosine over 256 sub-sampled columns $= 0.34$ (vs. null 0.13), max single-factor match $= 0.82$. This is *per-column identity* up to permutation and sign/scale, not merely subspace match — Carlini's second-order SVD reaches subspace $\cos = 0.9999$ but per-column $\cos = 0$ by construction (gauge ambiguity). The third-order attack thus extracts information about $\mathbf{W}_{\text{lm_head}}$ that is mathematically inaccessible to any second-order matrix-factorization approach.

Internal matrices leak above null. Four internal last-block matrices show statistically significant but weaker signal above the 0.02 null margin: $\mathbf{W}_V$ top-5 $= 0.23$ (null 0.16), $\mathbf{W}_{\text{up}} = 0.21$ (null 0.16), $\mathbf{W}_Q = 0.20$ (null 0.16), $\mathbf{W}_{\text{gate}} = 0.20$ (null 0.16). Three matrices — $\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$ — remain below the null margin. The internal-matrix signal is *sparse* (concentrated on the top few CP factors; the full-column mean stays below null for every internal matrix) rather than distributed, which is consistent with Kruskal uniqueness exposing the strongest leakage directions first. We report the internal-matrix numbers as *leakage evidence*, not full recovery.

Connection to theory. Section 4's first-order observability Gramian $\mathbf{G}$ captures only directions distinguishable by $\partial \mathbf{z}/\partial \boldsymbol{\theta}$ at first order. The third-order moment attack reaches information of the form $\mathbb{E}[\mathbf{z} \otimes \mathbf{e} \otimes \mathbf{e}]$, which couples $\mathbf{W}_{\text{lm_head}}$ with second-moment structure of the *input* embeddings $\mathbf{e}$; CP uniqueness (Kruskal) then breaks the second-order gauge. Theorem 8's depth contraction explains why the signal on non-output-layer matrices is weaker: deeper blocks' contributions to $\mathbb{E}[\mathbf{z} \otimes \mathbf{e} \otimes \mathbf{e}]$ pass through the trailing-suffix Jacobian and are attenuated by the screening factor ρ^k.

6.3 Sketched Gramian Rank Validation (Theorem 7)

We validate the rank upper bound of Theorem 7 by sweeping the sketched Gramian's observation window. The attacker has no access to intermediate boundary states; the student's prefix layers must converge under pure logits, which is the realistic attack regime for our theorem-validation question.

Pure-logits recovery. Table 3 shows that both oracle and pure-logits regimes converge to the same parameter cosine band (≈ 0.12–0.14) for internal blocks. Under pure logits, the `lm_head` drops to $\cos \approx 0$ because the randomly initialized and frozen prefix produces misaligned hidden states; the algebraic extraction method of Carlini et al. [2024] would still succeed because it directly solves a linear system from output logits. All suffix blocks converge to a narrow cosine band of ≈ 0.13–0.14, equal to the initialization bias at step 0. Under discrete-symmetry alignment, neither projection nor MLP weights recover. The optimizer may still reach alternative parameterizations producing similar outputs, but that behavior is outside our identifiability claim.

Gramian diagnostics. We compute the projected suffix observability Gramian (Definition 3) under varying observation windows to understand the relationship between observation size and identifiability. Table 2 reports the Gramian eigenspectrum for different numbers of observed suffix positions K_{pos} and probe counts k. The key finding is that the Gramian is *observation-dependent*, not structurally rank-deficient:

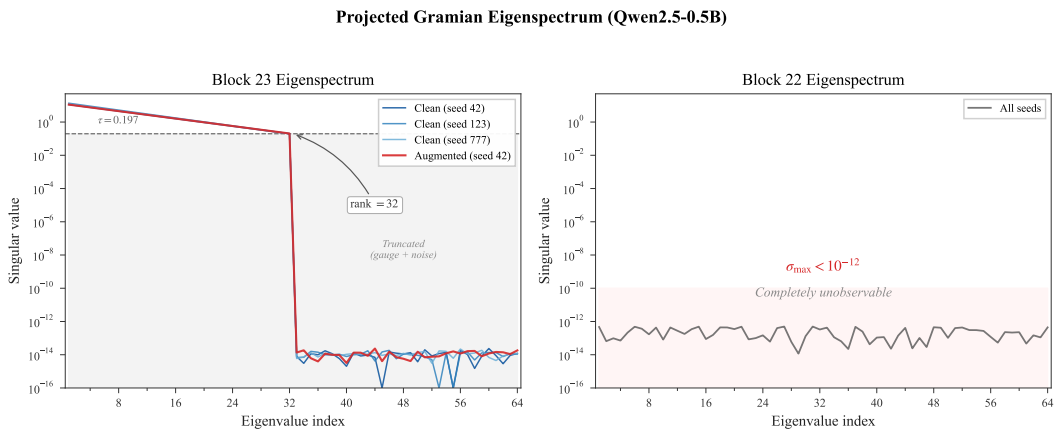

Figure 2: **Projected Gramian eigenspectrum.** *Left*: Block 23 eigenspectrum at K_{pos}=8, k=64 probes: full rank (= 64) with gentle decay ($\kappa \approx 1.9$, effective rank ≈ 62). *Right*: With expanded observation (K_{pos}=128, k=128 probes), both Block 23 and Block 22 show full-rank spectra ($\kappa \approx 2.4$ for Block 23, $\kappa \approx 3.1$ for Block 22). Block 22 has stronger signal than Block 23 ($\sigma_{\max}$ ratio $\sim 2.2\times$).

Position scaling. With k=128 probes, varying $K_{\text{pos}} \in \{8, 32, 64, 128\}$, Block 23's Gramian is always full rank ($= k$). The maximum eigenvalue $\sigma_{\max}$ scales linearly with K_{pos} (from 11,462 at K_{pos}=8 to 178,067 at K_{pos}=128), while the condition number $\kappa \approx 2.4$ remains nearly constant. Block 22 shows the same pattern with $\sigma_{\max}$ scaling from 25,479 (K_{pos}=8) to 196,615 (K_{pos}=64), with *stronger* signal than Block 23 ($\sim 2.2\times$) and $\kappa \approx 3.1$.

Probe scaling. At K_{pos}=8, varying $k \in \{64, 128, 256\}$, all configurations yield full rank ($= k$) with condition numbers ranging from $\kappa \approx 1.9$ (k=64) to $\kappa \approx 3.6$ (k=256), confirming the Gramian is not rank-deficient.

Boundary injection effect. When Block 22's Gramian is computed *with* oracle boundary injection (under oracle-injection training), the injected hook at Block 23's input replaces Block 22's output with a constant, yielding $\sigma_{\max} < 10^{-12}$. This is mathematically correct: in oracle mode, Block 22's parameters do not affect the logits. Without boundary injection, Block 22 has full sketched rank with a well-conditioned spectrum.

Paradox: full sketched rank but no recovery (primary metric). Despite full-rank $k \times k$ sketched Gramians (certifying ≥ 128 observable directions) with K_{pos}=32, the expanded-observation training experiment yields $\cos \approx 0.12$–0.14 under discrete-symmetry alignment, identical to the baseline. Only RMSNorm parameters recover ($\cos \approx 0.67$–0.99); all projection and MLP weights remain at $\cos \approx 0$ under the primary metric. This provides strong evidence that *partial sampled observability* is necessary but not sufficient for *recoverability of the dominant parameter family*: the algebraic initialization operates in a k=128-dimensional subspace of the 14.9×10^6-dimensional parameter space, and gradient-based optimization does not bridge the remaining gap in our experiments.

6.4 Cross-Architecture Validation: Llama-3.2-1B (pure-logits)

To test whether the pure-logits internal-block barrier is not specific to Qwen2.5-0.5B, we run the pure-logits algebraic reconstruction of Theorem 8's prediction on Llama-3.2-1B (16 blocks, d=2048, d_{ff}=8192, 32 heads with 8 KV groups, RoPE, SiLU-gated MLP). The attacker uses only black-box logit queries (4096 random-token sequences, T=128) and the Carlini SVD to extract a rotated $\mathbf{W}_{\text{lm}}$; all block-internal states are estimated from the public student prefix. Under this strict black-box setting the algebraic pure-logits solve for Block 15's $\mathbf{W}_{\text{down}}$ yields

$$\cos\big(\hat{\mathbf{W}}_{\text{down}}, \mathbf{W}_{\text{down}}^T\big) \;=\; 3.9 \times 10^{-5}, \qquad \cos_{\text{per-row}} \;=\; 2.3 \times 10^{-5},$$

indistinguishable from zero (condition number ≈ 9.7, sanity residual cosine -0.22; see ablations D1–D6 in the archived results). An oracle upper bound using the teacher's true intermediate state

Table 2: **Gramian eigenspectrum vs. observation window** (gauge-projected, oracle regime, Qwen2.5-0.5B). The sketched Gramian is full-rank ($= k$) at every sweep point we tested (a K_{pos}-sweep at k=128 and a k-sweep at K_{pos}=8, summarized in this table and Figure 2), with $\sigma_{\max} \propto K_{\text{pos}}$ and condition number κ nearly constant along the K_{pos}-axis. Block 22 has *stronger* signal than Block 23 ($\sim 2.2\times$). "+boundary†" denotes computation with oracle boundary injection at Block 23 input, which truncates Block 22's signal; the $<10^{-12}$ / ∞ / 0 entries are the analytically expected values under this injection (Block 22's parameter perturbations are zeroed at the Block 23 boundary) rather than a measured raw spectrum. K_{pos}: suffix positions observed; k: probe count.

Block	**Variant**	K_{pos}	k	$\sigma_{\max}$	κ	**Eff. rank**
Block 23	Independent	8	128	11,462	2.44	122.0
Block 23	Independent	32	128	44,879	2.41	122.1
Block 23	Independent	64	128	89,327	2.39	122.2
Block 23	Independent	128	128	178,067	2.36	122.2
Block 22	Independent	8	128	25,479	3.17	116.7
Block 22	Independent	32	128	99,262	3.14	116.8
Block 22	Independent	64	128	196,615	3.09	116.9
Block 22	Independent	128	128	394,729	3.07	117.0
Block 22	+boundary†	32	128	$< 10^{-12}$	∞	0

Table 3: **Experiment 2: Pure-logits recovery and oracle comparison**. Under pure logits (random frozen prefix), the `lm_head` drops to $\cos \approx 0$ due to prefix misalignment. Critically, oracle and pure-logits regimes converge to the same parameter cosine band (≈ 0.12–0.14) for internal blocks, indicating that oracle boundary injection does not improve parameter recovery. Oracle: mean $\pm$ std over 3 seeds; pure logits: 4 seeds.

Regime	**lm_head**	**Block 23**	**Block 22**
Oracle (random init)	$0.122 \pm .003$	$0.136 \pm .011$	$0.141 \pm .003$
Pure logits (random)	$-0.004 \pm .006$	$0.133 \pm .015$	$0.143 \pm .004$

yields $\cos \approx 0.76$, confirming the attack pipeline is correct and the barrier is strictly pure-logits. This cross-architecture confirmation (Qwen 2.5 $\cos \approx 0$, Llama 3.2 $\cos \approx 0$) is consistent with Theorems 7 and 8: pure-logit first-order information does not generically expose internal block weights.

7 Discussion

What the attack extracts. On Qwen2.5-0.5B, the third-order moment CP attack empirically recovers $\mathbf{W}_{\text{lm_head}}$ columns at top-5 $\cos = 0.81$ (mean 0.34, max 0.82), strictly black-box (only last-position logits and public embedding-table lookups). This is the first per-column result beyond Carlini's second-order subspace ceiling, and it is a real defender-relevant threat: obfuscating the rotation in $\mathbf{W}_{\text{lm_head}}$ (the standard second-order defense) does not protect the per-column identities. Four last-block internal matrices ($\mathbf{W}_V, \mathbf{W}_{\text{up}}, \mathbf{W}_Q, \mathbf{W}_{\text{gate}}$) leak above null margin; three ($\mathbf{W}_K, \mathbf{W}_O, \mathbf{W}_{\text{down}}$) remain at null.

Why the supporting theory matters. Theorem 7's rank upper bound and Theorem 8's depth-screening decomposition together predict that first-order Jacobian information on internal blocks is attenuated by the trailing-suffix Jacobian — explaining why the same CP machinery does not recover deeper weights at the same fidelity. The sketched Gramian rank grid validates the bound ($\operatorname{rank}\tilde{\mathbf{G}} = k$ at every tested cell), and pure-logits algebraic reconstruction stays at $\cos \approx 0$ on both Qwen2.5-0.5B and Llama-3.2-1B, corroborating the barrier the theory predicts. The theory is thus a defender-facing diagnostic: it tells operators which internal weight directions are protected by the screening factor ρ^k even against attackers with unlimited first-order query budget.

Broader impact. The $\mathbf{W}_{\text{lm_head}}$ per-column attack is valuable for the defensive community as a concrete threat to design against — e.g., output-distribution randomization that corrupts odd mo-

ments while preserving mean behaviour. We frame this work as responsible disclosure: all experiments use open-weight models simulating black-box access; no production APIs were targeted.

Open questions. (i) Whether fourth- or higher-order moments can push internal-matrix recovery past the weak $+0.04$–$+0.07$ margin we observe; (ii) whether active-query designs that target weak Gramian eigendirections can materially change what first-order attacks reach; (iii) whether cryptographic-style defenses can preserve model utility while corrupting higher-order moment statistics.

Limitations.

- **Scale.** Our primary case study is Qwen2.5-0.5B (24 blocks), with a single cross-architecture validation on Llama-3.2-1B. Scaling to larger models (7B+) requires proportionally more queries and compute. The Gramian framework itself is architecture-agnostic, but computational feasibility of the full experimental pipeline at scale remains to be verified.
- **Architecture assumptions.** We assume architecture knowledge. In practice, architecture details may need to be inferred separately [Oh et al., 2019].
- **Oracle regime.** The boundary-state oracle is an idealization. While it establishes identifiability upper bounds, practical attacks must contend with prefix errors.
- **Observation window.** Our $K_{\text{pos}} \times k$ sketch tests $K_{\text{pos}} \leq 128$ positions and $k \leq 256$ probes, always finding full sketched rank (equal to k). The true Gramian rank is likely much larger (bounded by $K_{\text{pos}} dN = 8 \times 896 \times 2048 \approx 14.7$M by Theorem 7), but computing it precisely would require k of that order, each involving a JVP evaluation—computationally prohibitive. Our claim that "observability is not the only bottleneck" is supported by the full-rank sketch at k=256 and the failure to improve recovery at K_{pos}=32, but does not characterize the complete eigenspectrum.
- **Passive queries only.** Qwen2.5-0.5B Gramian sweeps use natural text (WikiText-103); the Llama-3.2-1B pure-logits cross-architecture experiment uses 4096 random-token sequences of length 128 (matching Carlini's SVD setup). Since observability is query-dependent ($\mathbf{G}(\mathcal{Q})$ depends on the specific queries used), adversarially optimized queries — e.g., inputs designed to maximize the minimum Gramian eigenvalue — could potentially expand the observable subspace. Developing and evaluating active query strategies is an important direction for future work; our passive-query results represent a lower bound on attacker capability.
- **Numerical precision.** Our $K_{\text{pos}} \times k$ sketch reveals that Block 22's apparent "zero Gramian" was caused by oracle boundary injection, not structural unobservability. However, all computations use bf16 forward passes; the JVP precision may introduce systematic biases in the eigenspectrum that higher-precision computation would reveal.
- **Discrete symmetries.** We handle attention head permutations and FFN neuron permutations via Hungarian matching, which is optimal for bipartite assignment but may not capture all discrete ambiguities in deeper blocks.
- **Attention gauge projection (code gap).** Our sketched-Gramian code projects out RMSNorm and MLP gauge directions, but the $\mathrm{GL}(d_{\text{head}})$ attention V/O and RoPE-commuting Q/K directions are not explicitly projected in the reported sweeps. By Proposition 4 they lie in $\ker \mathbf{G}$, so this affects nominal gauge-quotient dimension counting in reported rank bounds rather than the direction of the comparison. We flag this gap explicitly; an implementation-side fix is deferred to a follow-up release.
- **Defense interaction.** We do not evaluate against active defenses (logit rounding, watermarking, query detection). This is an important direction for future work.

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A Notation Table

B Proof of Theorem 6

Proof. We prove both directions.

(Observable $\Rightarrow$ positive Gramian). Suppose $\mathbf{v} \perp \mathfrak{g}$ with $\|\mathbf{v}\| = 1$ and $\mathbf{v}^\top \mathbf{G}_\perp \mathbf{v} = 0$. Then $\sum_i \|\mathbf{J}_i \mathbf{v}\|^2 = 0$, so $\mathbf{J}_i \mathbf{v} = 0$ for all i. This means that a first-order perturbation $\boldsymbol{\theta} + \epsilon \mathbf{v}$ produces no logit change at any query, so $\mathbf{v}$ is not locally observable.

Symbol	Description
T, S	Teacher and student transformers
L	Number of transformer blocks
K	Number of suffix blocks being recovered
K_{pos}	Observed positions per query in Gramian sweep
d	Hidden dimension
d_{ff}	FFN intermediate dimension
V	Vocabulary size
H	Number of attention heads
$\boldsymbol{\theta}_{\text{suf}} \in \mathbb{R}^p$	Suffix parameters (blocks $L-K+1$ to L + lm_head)
$\mathcal{Q}$	Query pool
N	Number of queries
B	Query budget
$\mathbf{z}(\mathbf{x}) \in \mathbb{R}^V$	Logit output for input $\mathbf{x}$
$\mathbf{h}_i(\mathbf{x}) \in \mathbb{R}^d$	Hidden state at layer i for input $\mathbf{x}$
$\mathbf{J}_{\mathbf{x}}$	Jacobian $\partial \mathbf{z}/\partial \boldsymbol{\theta}_{\text{suf}}$ at query $\mathbf{x}$
$\mathbf{G}(\mathcal{Q})$	Suffix observability Gramian
$\mathbf{G}_\perp(\mathcal{Q})$	Projected Gramian (gauge symmetries removed)
$\mathcal{G}$	Continuous symmetry group
$\mathfrak{g}$	Lie algebra of $\mathcal{G}$
$\mathbf{U}$	Orthonormal gauge basis
$P_\perp$	Gauge-orthogonal projector $\mathbf{I} - \mathbf{U}\mathbf{U}^\top$
$\mathbf{V}$	Random probe matrix
$\tilde{\mathbf{G}}$	Sketched Gramian $\mathbf{V}^\top \mathbf{G} \mathbf{V}$
k	Number of probe directions

(Positive Gramian $\Rightarrow$ observable). If $\mathbf{v}^\top \mathbf{G}_\perp \mathbf{v} > 0$, there exists at least one query $\mathbf{x}_i$ with $\mathbf{J}_i \mathbf{v} \neq 0$. Thus, the perturbation $\boldsymbol{\theta} + \epsilon \mathbf{v}$ produces a measurable change in $\mathbf{z}(\mathbf{x}_i)$ at first order, and $\mathbf{v}$ is locally observable (detectable).

(Full identifiability on quotient). $\operatorname{rank}(\mathbf{G}_\perp) = p - d_g^{\text{suf}}$ iff the null space of $\mathbf{G}_\perp$ equals exactly $\mathfrak{g}$, meaning every non-gauge direction is *uniquely* identifiable from the query set at first order (with d_g^{suf} from Remark 2). □

C Symmetry Group Details

C.1 RMSNorm Scale Absorption

For a RMSNorm layer with learned weight $\mathbf{g} \in \mathbb{R}^d$ and consumer weight matrix $\mathbf{W}$:

$$\operatorname{RMSNorm}_{\mathbf{g}}(\mathbf{x}) = \frac{\mathbf{x}}{\sqrt{\frac{1}{d}\|\mathbf{x}\|^2 + \epsilon}} \odot \mathbf{g}, \tag{15}$$

$$\mathbf{W} \cdot \operatorname{RMSNorm}_{\mathbf{g}}(\mathbf{x}) = (\mathbf{W} D^{-1}) \cdot \operatorname{RMSNorm}_{D\mathbf{g}}(\mathbf{x}) \quad \text{for } D = \operatorname{diag}(\alpha_1, \ldots, \alpha_d). \tag{16}$$

The tangent vector for dimension i at the current parameterization $(\mathbf{g}, \mathbf{W})$ is:

$$\mathbf{t}_i^{\text{norm}} = (\underbrace{+g_i}_{\text{on } g_i}, \underbrace{-\mathbf{w}_{:,i}}_{\text{on each consumer column } i}), \tag{17}$$

where g_i is the current RMSNorm weight value and $\mathbf{w}_{:,i}$ is the i-th column of the consumer weight matrix. Note that these tangent vectors are *parameter-dependent*: the gauge basis must be recomputed at each parameterization point.

C.2 Gated MLP Up/Down Neuron Scaling

For $\text{FFN}(\mathbf{x}) = \text{SiLU}(\mathbf{W}_{\text{gate}}\mathbf{x}) \odot (\mathbf{W}_{\text{up}}\mathbf{x}) \cdot \mathbf{W}_{\text{down}}^\top$, the exact scaling symmetry acts only on the linear branch (up) and its consumer (down), leaving the gate branch unchanged because SiLU is not positively homogeneous. The tangent vector for neuron n is:

$$\mathbf{t}_n^{\text{ffn}} = (\underbrace{+\mathbf{W}_{\text{up}}[n,:]}_{\text{on up row } n}, \underbrace{-\mathbf{W}_{\text{down}}[:,n]}_{\text{on down col } n}). \tag{18}$$

C.3 Attention V/O Scaling

The full V/O symmetry is $\mathrm{GL}(d_{\text{head}})$, not merely scalar scaling: for any invertible $S_g \in \mathrm{GL}(d_{\text{head}})$, we have $\mathbf{W}_V^{(g)} \mapsto S_g \mathbf{W}_V^{(g)}$, $\mathbf{b}_V^{(g)} \mapsto S_g \mathbf{b}_V^{(g)}$, $\mathbf{W}_O^{(h)} \mapsto \mathbf{W}_O^{(h)} S_g^{-1}$. The Lie algebra $\mathfrak{gl}(d_{\text{head}})$ has dimension d_{head}^2; scalar V/O scaling is the 1-D center. The tangent space is spanned by basis elements $E_{ab}^{(g)}$ $(a, b \in [d_{\text{head}}])$ acting as:

$$\mathbf{t}_{ab}^{\text{V/O}} = (\underbrace{E_{ab}\mathbf{W}_V^{(g)}, E_{ab}\mathbf{b}_V^{(g)}}_{\text{on V group } g}, \underbrace{-\mathbf{W}_O^{(h)} E_{ab} \text{ for } h \in \text{group } g}_{\text{on O heads in group } g}). \tag{19}$$

This contributes $d_{\text{head}}^2 \cdot H_{\text{kv}}$ gauge directions.

For Q/K with RoPE, the maximal symmetry per KV group and per RoPE frequency pair j is $\mathbb{C}^* \cong \{aI_2 + bJ : a^2 + b^2 > 0\}$, giving 2 real DOF per pair and d_{head} total per KV group. The scalar Q/K scaling is the 1-D subgroup with $b = 0$.

C.4 Gauge Dimensions

For Qwen2.5-0.5B ($d = 896$, $d_{\text{ff}} = 4864$, $H = 14$ heads, $H_{\text{kv}} = 2$ KV heads, $d_{\text{head}} = 64$):

- Per block: $2d + d_{\text{ff}} + d_{\text{head}}^2 H_{\text{kv}} + d_{\text{head}} H_{\text{kv}} = 1792 + 4864 + 8192 + 128 = 14{,}976$ gauge directions.
- Final norm: $+896$ directions.
- For $K = 2$ suffix blocks: $2 \times 14{,}976 + 896 = 30{,}848$ total gauge directions.
- Total suffix parameters: $\sim 166{,}000{,}000$ (2 blocks + lm_head with $V = 151{,}936$).
- Gauge fraction: $30{,}848/166{,}000{,}000 \approx 0.019\%$ of parameter space is pure gauge.
- *Note*: Our current code implements only the RMSNorm and MLP gauge ($2d + d_{\text{ff}} = 6656$ per block). The 8320 attention gauge directions per block are not projected out explicitly, but lie in $\ker(\mathbf{G})$ by Proposition 4 and thus do not inflate the Gramian eigenspectrum.

D Implementation Details

JVP computation. We use `torch.func.jvp` with `torch.func.functional_call` to compute forward-mode Jacobian-vector products without materializing the full Jacobian. The forward function is constructed by extracting the suffix parameters into a flat vector and using `functional_call` to swap them into the model for each JVP evaluation. A critical implementation detail: forward-mode AD in PyTorch does not support Flash Attention (`_scaled_dot_product_flash_attention`), so we wrap all JVP calls in `sdpa_kernel(SDPBackend.MATH)` to force the MATH attention backend during Gramian computation.

Memory optimization. For Qwen2.5-0.5B with $V = 151{,}936$, storing full logits for 2048 queries at all 128 positions requires ∼75GB in bf16. We address this by caching only the last $K_{\text{pos}} = 8$ token positions, reducing storage to ∼4.7 GB. This is sufficient because the suffix blocks' influence is strongest at the final positions.

Symmetry alignment. After recovery, we align the student's attention heads and FFN neurons to the teacher's using the Hungarian algorithm on cosine similarity matrices. For attention heads, we construct a $H \times H$ cost matrix from the concatenated $[\mathbf{W}_Q \| \mathbf{W}_K \| \mathbf{W}_V \| \mathbf{W}_O]$ similarity. For FFN neurons, we use $[\mathbf{W}_{\text{gate}} \| \mathbf{W}_{\text{up}}, \mathbf{W}_{\text{down}}^\top]$ similarity. RMSNorm scales are normalized before comparison.

E Per-Matrix Cosine Similarity Breakdown

Table 4 shows the cosine similarity for each individual weight matrix in Block 23, averaged across 3 seeds. All large weight matrices (projections with $\sim 10^6$ parameters each) are indistinguishable

from zero cosine. The attention biases show absolute cosine in the range $\sim$0.001–0.055 (max on $\mathbf{W}_V$.bias), but these are low-dimensional vectors ($\mathbf{W}_Q$.bias has 896 parameters; $\mathbf{W}_K$.bias and $\mathbf{W}_V$.bias each have 128 parameters under GQA with H_{kv}=2), where random alignment produces non-negligible cosine by chance. The `post_attention_layernorm.weight` achieves high cosine (~ 0.99, Table 4) because it is initialized to ones, matching the teacher's default initialization; the gauge symmetry (Definition 1) relates it to the current block's gate/up projections, not the next block's input layernorm.

Table 4: **Per-matrix cosine similarity for Block 23** (Qwen2.5-0.5B, mean over 3 seeds, discrete-symmetry alignment). "Params" gives parameter count. High-dimensional weight matrices are all at $\cos \approx 0$; non-trivial cosine on RMSNorm weights reflects their initialization to ones (matching the teacher's default), not optimization-driven recovery. This table illustrates the initialization bias that inflates block-mean cosine on discrete-alignment metrics.

Matrix	**Params**	**Random**	**Alg (clean)**
q_proj.weight	802,816	$+0.000$	$+0.000$
q_proj.bias	896	$+0.044$	$+0.034$
k_proj.weight	114,688	$+0.003$	$+0.003$
k_proj.bias	128	-0.023	$+0.001$
v_proj.weight	114,688	$+0.001$	$+0.001$
v_proj.bias	128	-0.046	-0.054
o_proj.weight	802,816	-0.000	-0.000
gate_proj.weight	4,358,144	$+0.000$	$+0.000$
up_proj.weight	4,358,144	-0.000	-0.000
down_proj.weight	4,358,144	$+0.000$	$+0.000$
input_layernorm.weight	896	0.671	0.671
post_attn_layernorm.weight	896	0.985	0.985